

SECURITY IN COMPUTING – COMPLETE MASTER NOTES

M.Sc. Computer Science (Software)

UNIT I – COMPUTER SECURITY FUNDAMENTALS

1. Introduction to Computer Security

Meaning of Computer Security

Computer security refers to the protection of computer systems, software, hardware, networks, and digital information from unauthorized access, misuse, modification, destruction, or theft. The main objective of computer security is to ensure that information and systems remain safe and function properly.

Today almost every organization depends on computer systems. Banks store customer information digitally, hospitals maintain medical records online, and universities manage student data through databases. Because important information is stored electronically, strong security measures are necessary.

Computer security protects:

- Personal information
- Financial records
- Government data
- Business information
- Software and hardware resources

Without proper security, attackers can:

- Steal confidential information
 - Damage systems
 - Interrupt services
 - Misuse resources
 - Cause financial losses
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2. Objectives of Computer Security (CIA Triad)

The three main objectives of computer security are known as the CIA Triad.

A. Confidentiality

Confidentiality means protecting information from unauthorized access. Only authorized users should be able to access confidential information.

Examples

- Bank passwords
- ATM PINs
- Medical records
- Aadhaar numbers

Methods used for confidentiality

1. Password protection
2. Encryption
3. Authentication systems
4. Access control
5. Biometric systems

Importance of confidentiality

- Maintains privacy
- Prevents data theft
- Protects sensitive information

Problems caused when confidentiality is lost

- Identity theft
 - Financial fraud
 - Leakage of private data
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B. Integrity

Integrity means maintaining the accuracy, consistency, and reliability of data. Information should not be modified by unauthorized persons.

Example

Student marks stored in a university database should not be changed illegally.

Methods used to maintain integrity

1. Hashing
2. Digital signatures
3. File permissions
4. Access control
5. Checksums

Importance of integrity

- Ensures trustworthiness of information
- Prevents unauthorized changes
- Maintains data accuracy

Problems caused by loss of integrity

- Incorrect records
 - Financial losses
 - System failures
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C. Availability

Availability means systems and information should remain accessible whenever required by authorized users.

Example

Online banking websites should remain available continuously.

Methods used for availability

1. Backup systems
2. Disaster recovery plans
3. UPS systems
4. Redundant servers
5. Load balancing

Importance of availability

- Ensures uninterrupted services
- Improves customer trust
- Supports business operations

Threats to availability

- Denial of Service attacks
 - Hardware failures
 - Natural disasters
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3. Need for Computer Security

Computer security is important because:

1. Cyber attacks are increasing rapidly.
2. Organizations store sensitive information digitally.
3. Online transactions require protection.

4. Systems must be protected against hackers and malware.
 5. Security maintains privacy and trust.
 6. Security helps avoid financial losses.
 7. Government and business data require protection.
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4. Threats, Attacks and Assets

Asset

An asset is anything valuable that requires protection.

Examples of assets

- Hardware
 - Software
 - Databases
 - Passwords
 - Financial records
 - Network infrastructure
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Threat

A threat is any possible danger that may harm a computer system.

Types of threats

1. Human threats
2. Natural threats
3. Environmental threats

Examples

- Hackers
 - Viruses
 - Fire
 - Flood
 - Power failure
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Attack

An attack is an attempt to exploit system vulnerabilities.

Objectives of attacks

- Steal information
- Damage systems

- Interrupt services
 - Gain unauthorized access
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5. Types of Attacks

A. Passive Attacks

In passive attacks, attackers monitor information without changing it.

Characteristics

- Difficult to detect
- Does not modify data
- Mainly affects confidentiality

Examples

1. Eavesdropping
2. Traffic analysis

Prevention methods

- Encryption
 - Secure communication protocols
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B. Active Attacks

In active attacks, attackers modify, damage, or destroy information.

Characteristics

- Easier to detect
- Affects integrity and availability

Examples

1. Malware attacks
2. Message modification
3. Denial of Service attacks
4. Masquerade attacks

Prevention methods

- Firewalls
 - IDS
 - Authentication
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6. Security Functional Requirements

A. Authentication

Authentication verifies user identity.

Methods of authentication

1. Username and password
 2. OTP
 3. Smart cards
 4. Biometrics
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B. Authorization

Authorization determines user permissions after login.

Example

An admin can modify records while a student can only view records.

C. Accountability

Accountability tracks user activities.

Example

System logs store login records and user actions.

D. Privacy

Privacy means protecting personal information from misuse.

E. Non-Repudiation

Non-repudiation ensures that users cannot deny their actions.

Example

Digital signatures.

7. Security Architecture

Security architecture is the structure used to provide protection for systems and information.

Components of security architecture

1. Firewalls
 2. Antivirus software
 3. Encryption systems
 4. Authentication systems
 5. IDS and IPS
 6. Access control systems
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8. Scope of Computer Security

Computer security covers: 1. Hardware security 2. Software security 3. Network security 4. Database security 5. Internet security 6. Mobile security 7. Cloud security 8. Web security

9. Security Trends and Strategies

Modern security trends

1. Cloud security
 2. Artificial intelligence in security
 3. Multi-factor authentication
 4. Zero trust architecture
 5. Blockchain security
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Security strategies

1. Strong password policies
 2. Regular software updates
 3. Security awareness training
 4. Data encryption
 5. Backup systems
 6. Security audits
 7. Access control
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10. Cryptography

Meaning of Cryptography

Cryptography is the science of securing information by converting readable information into unreadable form.

It protects:

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

Applications of cryptography

1. Online banking
 2. ATM systems
 3. E-commerce
 4. Secure communication
 5. Digital signatures
-

11. Cryptography Terminology

Plain Text

Original readable message.

Cipher Text

Encrypted unreadable message.

Encryption

Process of converting plain text into cipher text.

Decryption

Process of converting cipher text back into readable form.

Key

Secret value used during encryption and decryption.

12. Types of Cryptography

A. Symmetric Key Cryptography

Uses the same key for encryption and decryption.

Advantages

1. Fast
2. Efficient for large data
3. Simple implementation

Disadvantages

1. Key sharing problem
2. Less secure for communication

Examples

- AES
 - DES
-

B. Asymmetric Key Cryptography

Uses two keys: 1. Public key 2. Private key

Advantages

1. More secure
2. Solves key distribution problem

Disadvantages

1. Slower processing
2. Complex implementation

Example

RSA

13. Classical Encryption Techniques

A. Substitution Cipher

In substitution cipher, characters are replaced by other characters.

Caesar Cipher

Each letter shifts by a fixed number.

Advantages:

- Simple implementation

Disadvantages:

- Easy to break
-

B. Transposition Cipher

Characters remain the same but positions are changed.

Example

HELLO → LEHOL

Advantages:

- Easy to use

Disadvantages:

- Weak security
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14. Cryptanalysis

Cryptanalysis is the process of breaking encrypted messages without knowing the key.

Methods of cryptanalysis

1. Brute force attack
 2. Frequency analysis
 3. Known plaintext attack
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15. Program Security

Program security protects software from vulnerabilities and attacks.

Features of secure programs

1. Prevent unauthorized access
 2. Handle errors safely
 3. Protect user data
 4. Avoid vulnerabilities
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16. Non-Malicious Program Errors

These are accidental programming mistakes.

Examples

1. Logic errors
 2. Buffer overflow
 3. Memory leaks
 4. Input validation errors
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17. Viruses and Malicious Code

Virus

A virus is a malicious program that infects files and spreads.

Effects of viruses

1. Slow system performance
 2. File corruption
 3. Data deletion
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Worm

A worm spreads automatically through networks.

Trojan Horse

A Trojan looks useful but secretly performs harmful activities.

Spyware

Spyware secretly collects user information.

Ransomware

Ransomware locks files and demands money.

18. Control Against Program Threats

Methods

1. Antivirus software
2. Firewalls
3. User awareness
4. Software updates
5. Backup systems
6. Access control

UNIT II – DATABASE SECURITY AND NETWORK SECURITY

1. Database Security

Meaning of Database Security

Database security refers to the protection of databases from unauthorized access, misuse, corruption, or theft.

Databases contain important information such as:

- Customer records
- Banking details
- Employee information
- Student data

Therefore database security is extremely important.

2. Importance of Database Security

1. Protects confidential data
2. Prevents unauthorized access
3. Maintains integrity of records
4. Prevents data theft

5. Ensures availability of information
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3. Database Management System (DBMS)

A DBMS is software used to store, manage, and organize data.

Functions of DBMS

1. Data storage
2. Data retrieval
3. Data management
4. Security management

Examples

1. MySQL
 2. Oracle
 3. SQL Server
 4. MongoDB
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4. Relational Databases

Relational databases store data in tables.

Features

1. Rows and columns
 2. Relationships between tables
 3. SQL support
-

5. Database Access Control

Database access control restricts access to authorized users.

Types of access control

A. Discretionary Access Control (DAC)

Owner controls permissions.

B. Mandatory Access Control (MAC)

Access based on security levels.

C. Role-Based Access Control (RBAC)

Permissions based on user roles.

Example:

- Admin
 - Teacher
 - Student
-

6. Inference Problem

Inference occurs when attackers guess confidential information using available data.

Example

A user may guess salary details from average department salary information.

7. Reliability and Integrity

Reliability

Database should function correctly without failure.

Integrity

Data should remain accurate and consistent.

8. Sensitive Data

Sensitive data requires strong protection.

Examples

1. Passwords
 2. Medical records
 3. Aadhaar numbers
 4. Financial details
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9. Database Encryption

Database encryption converts information into unreadable form.

Advantages

1. Prevents theft
 2. Protects confidentiality
 3. Improves security
-

10. Network Security

Meaning

Network security protects networks from unauthorized access and attacks.

11. Network Threats

Common threats

1. Malware
 2. Phishing
 3. Spoofing
 4. Denial of Service attacks
 5. Eavesdropping
 6. Man-in-the-middle attacks
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12. Security Controls

Types of controls

1. Firewalls
 2. Antivirus software
 3. Encryption
 4. VPN
 5. IDS
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13. Firewall

Meaning

A firewall is a security device that monitors and filters network traffic.

Need for firewall

1. Blocks unauthorized access
 2. Protects networks
 3. Monitors traffic
 4. Prevents attacks
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Characteristics of firewall

1. Packet filtering
 2. Access control
 3. Monitoring
 4. Logging
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14. Types of Firewalls

A. Packet Filtering Firewall

Checks packets according to predefined rules.

Advantages

1. Fast
2. Simple

Disadvantages

1. Limited security
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B. Proxy Firewall

Acts as an intermediary between users and internet.

Advantages

1. Better security
2. Hides internal network

Disadvantages

1. Slower performance
-

C. Stateful Inspection Firewall

Tracks active sessions and connections.

Advantages

1. Better monitoring
 2. Improved protection
-

15. Firewall Basing

Firewall basing means proper placement of firewall within a network.

16. Intrusion Detection System (IDS)

Meaning

IDS monitors suspicious activities and detects attacks.

17. Types of IDS

A. Network IDS

Monitors network traffic.

B. Host IDS

Monitors activities inside a computer.

18. Goals of IDS

1. Detect attacks
 2. Alert administrators
 3. Monitor activities
 4. Improve security
-

19. Strengths of IDS

1. Early attack detection
 2. Better monitoring
 3. Security analysis
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20. Limitations of IDS

1. False alarms
 2. Cannot stop all attacks
 3. Complex management
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UNIT III – INTERNET SECURITY AND OPERATING SYSTEM SECURITY

1. Internet Security

Internet security protects online communication, data, and systems from cyber threats.

Importance

1. Secure online transactions
 2. Protect user privacy
 3. Prevent cyber attacks
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2. SSL (Secure Socket Layer)

SSL is a security protocol used for encrypted communication.

Features

1. Encryption
2. Authentication
3. Data integrity

Applications

1. Online banking
2. E-commerce
3. HTTPS websites

3. TLS (Transport Layer Security)

TLS is an improved version of SSL.

Advantages of TLS

1. Stronger encryption
 2. Better security
 3. Faster performance
-

4. Difference Between SSL and TLS

SSL

1. Older protocol
2. Less secure
3. Slower

TLS

1. Newer protocol
 2. More secure
 3. Faster
-

5. IPv4 and IPv6 Security

IPv6 provides improved security features compared to IPv4.

Features of IPv6 security

1. Better authentication
 2. Improved encryption
 3. IPSec support
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6. Kerberos

Kerberos is a network authentication protocol using tickets.

Advantages

1. Strong authentication

2. Password protection
 3. Prevents unauthorized access
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7. X.509 Certificate

X.509 is a digital certificate used for identity verification.

Contains

1. Owner information
 2. Public key
 3. Digital signature
 4. Certificate authority details
-

8. Public Key Infrastructure (PKI)

PKI manages digital certificates and encryption keys.

Components of PKI

1. Certificate Authority
2. Registration Authority
3. Public key
4. Private key
5. Digital certificates

Advantages of PKI

1. Strong authentication
 2. Secure communication
 3. Digital signatures
-

9. Linux Security Model

Linux security is based on users, permissions, and access control.

Features

1. User accounts
 2. File permissions
 3. Access control
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10. File System Security

Linux file permissions: 1. Read 2. Write 3. Execute

11. Linux Vulnerabilities

Causes

1. Weak passwords
 2. Outdated software
 3. Improper configuration
-

12. Linux System Hardening

Methods

1. Disable unused services
 2. Install updates
 3. Use firewall
 4. Strong passwords
 5. Access control
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13. Windows Security Architecture

Windows provides security through: 1. User Account Control 2. Windows Defender 3. NTFS permissions 4. Firewall

14. Windows Vulnerabilities

Examples

1. Malware attacks
 2. Weak passwords
 3. Outdated systems
-

15. Windows Security Defense

Methods

1. Antivirus software
 2. Firewall
 3. Automatic updates
 4. Security patches
-

16. Browser Defenses

Modern browsers provide: 1. Safe browsing 2. Pop-up blocking 3. Website warnings 4. Download protection

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UNIT IV – PHYSICAL SECURITY AND ETHICAL ISSUES

1. Physical Security

Physical security protects hardware, buildings, and infrastructure from physical damage.

2. Physical Security Threats

Examples

1. Fire
 2. Flood
 3. Theft
 4. Earthquake
 5. Power failure
 6. Vandalism
-

3. Prevention and Mitigation Measures

Methods

1. CCTV cameras
2. Security guards

3. Fire alarms
 4. UPS systems
 5. Biometric systems
 6. Access control systems
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4. Disaster Recovery

Disaster recovery is the process of restoring systems after disasters.

Steps in disaster recovery

1. Backup creation
2. Recovery planning
3. System restoration
4. Testing recovery plans

Importance

1. Maintains availability
 2. Reduces downtime
 3. Prevents data loss
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5. Security Breaches

A security breach occurs when unauthorized users gain access to systems or information.

Causes

1. Weak passwords
 2. Malware attacks
 3. Human error
 4. Poor security policies
-

6. Security Auditing Architecture

Security auditing examines security controls and activities.

Importance

1. Identifies weaknesses
 2. Improves security
 3. Ensures compliance
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7. Security Audit Trail

Audit trail is the record of user activities and system events.

Importance

1. Monitoring
 2. Investigation
 3. Accountability
-

8. Security Risk Assessment

Risk assessment identifies threats and vulnerabilities.

Steps

1. Identify threats
 2. Analyze vulnerabilities
 3. Estimate impact
 4. Apply controls
-

9. Security Controls and Safeguards

A. Preventive Controls

Prevent attacks before they occur.

Examples

1. Passwords
 2. Firewalls
 3. Access control
-

B. Detective Controls

Detect attacks after they occur.

Examples

1. IDS
 2. Security logs
 3. Monitoring systems
-

C. Corrective Controls

Recover systems after attacks.

Examples

1. Backup restoration
 2. Recovery systems
 3. Disaster recovery plans
-

10. IT Security Plan

An IT security plan contains security policies and procedures.

Components

1. Security policies
 2. Risk management
 3. Access control
 4. Incident response
-

11. Cybercrime and Computer Crime

Cybercrime refers to illegal activities performed using computers.

Examples

1. Hacking
 2. Identity theft
 3. Online fraud
 4. Cyber stalking
 5. Phishing
 6. Software piracy
-

12. Intellectual Property

Intellectual property protects creations such as software, books, music, and inventions.

Types

1. Copyright
2. Patent
3. Trademark

13. Privacy

Privacy means protecting personal information from misuse.

Importance

1. Protects personal information
 2. Prevents identity theft
 3. Maintains trust
-

14. Ethical Issues

Ethical issues in computing refer to the moral principles and standards that should be followed while using computers, networks, and digital technologies. With the rapid growth of the internet and information technology, ethical behavior has become extremely important. Ethical use of technology ensures that computer systems and digital resources are used responsibly and legally.

In the modern world, computers are used in schools, banks, hospitals, businesses, and government organizations. Misuse of computers can create serious problems such as data theft, hacking, software piracy, privacy violations, and cybercrime. Therefore, users must follow proper ethical standards while using technology.

Importance of Ethics in Computing

1. Protects privacy of individuals.
 2. Prevents misuse of technology.
 3. Encourages responsible behavior.
 4. Maintains trust in digital systems.
 5. Reduces cybercrime and illegal activities.
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Common Ethical Issues in Computing

1. Privacy Violation

Privacy violation occurs when personal information is collected, accessed, or shared without permission. Many organizations store user data such as passwords, banking information, phone numbers, and medical records. Unauthorized access to such data is unethical.

Example

Stealing social media passwords or accessing someone's private emails without permission.

Prevention Methods

- Strong passwords
 - Data encryption
 - Access control
 - Privacy policies
-

2. Hacking

Hacking means gaining unauthorized access to computer systems or networks. Ethical hackers test systems legally to improve security, but malicious hackers illegally access systems for personal benefit.

Effects of Hacking

1. Data theft
 2. Financial losses
 3. System damage
 4. Privacy violations
-

3. Software Piracy

Software piracy means copying or using software illegally without permission from the owner.

Examples

- Using cracked software
- Downloading pirated applications
- Sharing paid software illegally

Effects

1. Financial loss to software companies
 2. Legal punishment
 3. Security risks due to malware
-

4. Intellectual Property Theft

Intellectual property includes software, music, books, inventions, and research work. Copying or using such work without permission is unethical.

Types of Intellectual Property Protection

1. Copyright
 2. Patent
 3. Trademark
-

5. Spreading False Information

Using computers or social media to spread fake news, rumors, or harmful information is unethical.

Effects

1. Public confusion
 2. Social problems
 3. Loss of trust
-

6. Cyberbullying and Online Harassment

Cyberbullying means using digital platforms to harass, threaten, or insult others.

Examples

- Posting harmful comments
 - Sending threatening messages
 - Online harassment through social media
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Ethical Practices in Computing

Users should follow ethical practices while using computers and the internet.

Important Ethical Practices

1. Respect privacy of others.
 2. Avoid hacking and illegal access.
 3. Use licensed software.
 4. Protect confidential information.
 5. Respect intellectual property rights.
 6. Use technology responsibly.
 7. Avoid spreading malware or harmful content.
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Conclusion

Security in computing is essential for protecting systems, networks, and information from cyber threats. Organizations and individuals must use proper security mechanisms such as encryption, authentication, firewalls, IDS, antivirus software, and security policies to maintain confidentiality, integrity, and availability of information. Along with technical security measures, ethical use of technology is also important. Proper security and ethical practices help organizations prevent cyber attacks, protect sensitive data, maintain user trust, and ensure safe use of digital technologies.

Security in computing is essential for protecting systems, networks, and information from cyber threats. Organizations must use strong security mechanisms such as encryption, authentication, firewalls, IDS, antivirus software, and security policies to maintain confidentiality, integrity, and availability of

information. Proper security practices help organizations prevent cyber attacks, maintain trust, and ensure smooth functioning of digital systems.